

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – VII
PAPER –2
TEST DATE: 07-08-2022

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

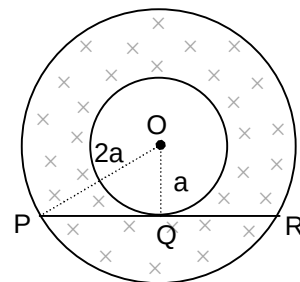
Physics

PART – I

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. A uniform magnetic field is applied in a hollow cylindrical region of inner radius a and outer radius $2a$. Magnetic field varies with time as $B = e^t$. A conducting wire is placed along chord horizontally. It has three points P, Q and R. Select correct option(s) at $t = 0$.

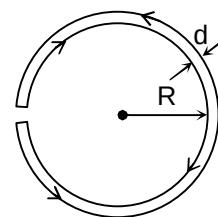


- (A) $V_P - V_Q = -\left(\frac{\sqrt{3}}{2} - \frac{\pi}{6}\right)a^2$
 (B) $V_Q - V_R = 0$
 (C) $V_P - V_R = \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6}\right)a^2$
 (D) $V_O - V_P = 0$

2. A copper sphere is given Q charge and kept isolated. Now temperature of sphere is increased by ΔT . Linear expansion coefficient of copper is α and specific heat capacity S . Assume the size change is only due to temperature.

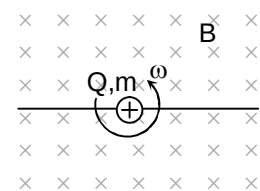
- (A) Change in electrostatic potential energy is $-\frac{Q^2\alpha\Delta T}{8\pi\epsilon_0 R}$
 (B) Change in electrostatic potential energy is $-\frac{Q^2\alpha\Delta T}{4\pi\epsilon_0 R}$
 (C) Total heat needed for this process is $\left(\frac{-Q^2\alpha}{8\pi\epsilon_0 R} + mS\right)\Delta T$
 (D) Total heat needed for this process is $\left(\frac{-Q^2\alpha}{4\pi\epsilon_0 R} + mS\right)\Delta T$

3. A conducting loop having large radius R is in form of two concentric rings separated by small distance d . Loop carries current i . Select correct option about magnetic field at centre.



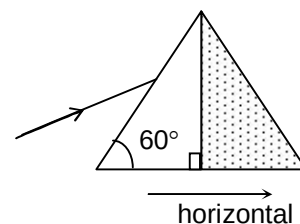
- (A) $B_{\text{center}} \propto i$
 (B) $B_{\text{center}} \propto \frac{1}{d}$
 (C) $B_{\text{center}} \propto \frac{1}{R}$
 (D) $B_{\text{center}} \propto \frac{1}{R^2}$

4. A charge Q having mass m can slide on a smooth rigid heavy rod. System is placed in uniform magnetic field B . Charge is initially at centre of rod. Rod is being rotated about center with uniform angular velocity ω as shown in the figure. (neglect gravity)



- (A) For stable equilibrium $\omega < \frac{qB}{m}$

- (B) For stable equilibrium $\omega < \frac{qB}{2m}$
- (C) Particle can perform oscillations but not SHM along the rod
- (D) Particle can perform SHM about center along the rod
5. An equilateral triangular prism is made using two materials of refractive index n and $n + \Delta n$, where $\Delta n \ll n$. A ray is sent into the prism such that it is horizontal inside the prism. Then,
- (A) incident angle of ray is independent of n .
- (B) incident angle of ray is independent of Δn .
- (C) Total deviation of ray increases with Δn .
- (D) Total deviation of ray is independent of Δn .
6. Living wood takes radioactive carbon from the atmosphere during the process of photo synthesis. The proportion of C^{14} and C^{12} being 8×10^{-14} . When wood dies C^{14} decays with half life 2.1×10^9 s. 4g of carbon from a piece of dead wood gave a total count of 20 disintegrations per minute. Then, (Take $N_A = 6 \times 10^{23}$, $\ln 2 = 0.7$)
- (A) Age of wood is 8.4×10^9 s.
- (B) Age of wood is 6.3×10^9 s.
- (C) Number of C^{14} at present in the wood piece is 10^9
- (D) Number of C^{14} at present in the wood piece is 10^{10}



Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

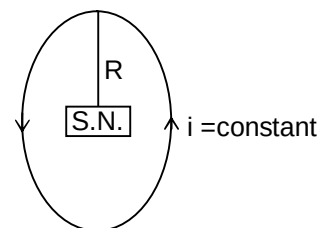
A small positive charge q is placed in an infinite viscous liquid. The charge is thrown with some initial speed in x - y plane and it is observed that it comes to rest after some time. Now a uniform magnetic field $\vec{B}$ applied along z -axis. Charge is again projected with some initial velocity. The displacements in two cases are $\vec{S}_1$ and $\vec{S}_2$. The viscous force due to medium is $\vec{F} = -b\vec{v}$. (where b is constant and $\vec{v}$ is velocity)

7. Angle between $\vec{S}_1$ and $\vec{S}_2$ is
- (A) $\sin^{-1}\left(\frac{2BqS_2}{bS_1}\right)$
- (B) $\sin^{-1}\left(\frac{BqS_2}{2bS_1}\right)$
- (C) $\sin^{-1}\left(\frac{BqS_2}{bS_1}\right)$
- (D) $\cos^{-1}\left(\frac{BqS_2}{bS_1}\right)$

8. If heat in case-I is H_1 and heat in case-II is H_2 , then $\frac{H_1}{H_2}$ is
- (A) 2
(B) $\frac{1}{2}$
(C) 3
(D) 1

Paragraph for Question Nos. 09 and 10

A small bar magnet placed along axis of a current carrying coil. Magnetic moment of magnet is $\vec{M}$ and current in ring is i . System is isolated in free space. Ring is kept fixed and mass of magnet is m . (The direction of magnetic field due to coil at its centre is parallel to $\vec{M}$)



9. Bar magnet is displaced along x-axis and releases. Time period for small oscillations is
- (A) $2\pi\sqrt{\frac{2mR^3}{3\mu_0 Mi}}$
(B) $\pi\sqrt{\frac{2mR^3}{3\mu_0 Mi}}$
(C) $\frac{\pi}{2}\sqrt{\frac{mR^3}{\mu_0 Mi}}$
(D) $\pi\sqrt{\frac{mR^3}{2\mu_0 Mi}}$
10. The escape velocity of bar magnet from centre of ring is
- (A) $\sqrt{\frac{M\mu_0 i}{3Rm}}$
(B) $\sqrt{\frac{M\mu_0 i}{2Rm}}$
(C) $\sqrt{\frac{M\mu_0 i}{Rm}}$
(D) $\sqrt{\frac{3M\mu_0 i}{Rm}}$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

11. Refractive index of a material depends on incidence angle such that refraction angle is always is $\frac{1}{3}$ times of incidence angle when ray is coming from air into medium. The error in refractive index will be minimum for incidence angle $\frac{\pi k}{8}$ radian. Find the value of k .

12. A rope of length 1 meter and variable linear mass density is suspended from one end vertically linear mass density of lower end is 1 kg/m^3 . When a pulse is given at lower end it travels uniformly at $\sqrt{10} \text{ m/s}$. Then the distance of center of mass of rope from lower end (in cm) is ($e = 2.71$)
13. In process of making value of gravitational constant 1, a new system is adopted in which unit of mass is 15 kg and unit of time is second. The new unit of length is k meter. Determine the value of 1000k. (take $G = \frac{20}{3} \times 10^{-11}$)

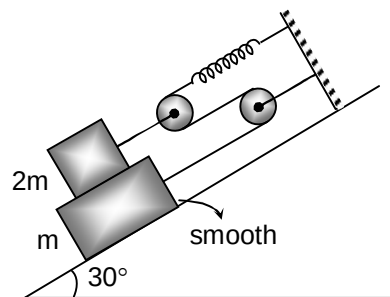
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

Two blocks are placed on an inclined plane. There is friction between blocks ($\mu = 0.5$) but plane is smooth. The pulley system has a spring ($k = 100 \text{ N/m}$) and system is initially at rest (take $m = 3 \text{ kg}$).



14. The time period of SHM for small oscillation in seconds is.....
15. Maximum frictional force between blocks for 1 mm amplitude in newton is.....

Question Stem for Question Nos. 16 and 17

Question Stem

A parallel plate capacitor is filled with a dielectric of variable dielectric constant from one plate to other as

$$k = 1 + \frac{x}{d}$$

Where x is distance from left plate in meters, d is distance between plates in meters.

16. A conducting plate is inserted parallel to plates at distance Nd from first plate such that two parallel plate capacitors can be formed of equal capacitance. Value of N is
17. Ratio of magnitudes of volume charge density at $x = \frac{d}{3}$ and $x = \frac{4d}{5}$ is $2k$. Find the value of k .

Question Stem for Question Nos. 18 and 19

Question Stem

A brick is thrown at an angle θ with respect to ground. Assume that the long face of brick remains parallel to ground at all times and there is no deformation in the ground or brick when it hits the ground. The coefficient of friction between brick and ground is μ . Ground is perfectly inelastic. The angle of projection to maximize total horizontal distance before coming to rest is X radian, if $\mu = \frac{1}{\sqrt{3}}$ and Y radian, if $\mu = \sqrt{3}$.

(take $\pi = 3.14$)

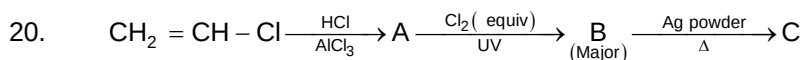
18. The value of X is
19. The value of Y is.....

Chemistry

PART – II

Section – A (Maximum Marks: 24)

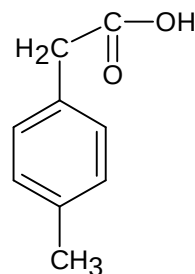
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).



How many of the following statements is/are correct about the above reaction sequences?

- (A) B has higher dipole moment than chloroform
- (B) C gives positive test with ammoniacal AgNO_3
- (C) A on hydrolysis produces a compound which gives positive iodoform test
- (D) C gives white ppt. on treatment with aq. AgNO_3

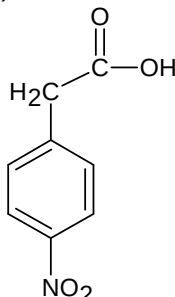
21. How many of the following decarboxylates more easily than



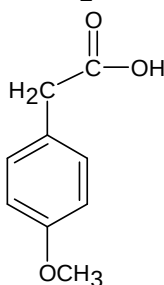
(when heated with

soda lime)?

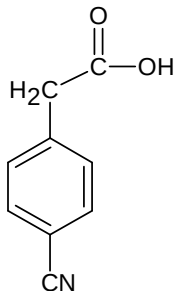
(A)

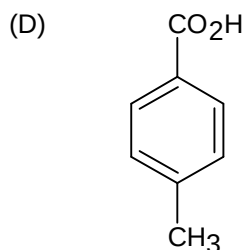


(B)

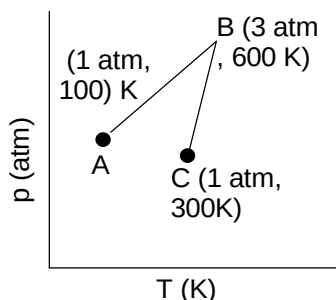


(C)





22. One mole of an ideal gas is subjected to a two step reversible process (A – B and B – C). The pressure at A and C is same. Mark the correct statement(s):



- (A) Work involved in the path AB is zero
 (B) In the path AB work will be done on the gas by the surroundings
 (C) Volume of gas at C = 3 × vol of gas at A
 (D) Vol. of gas at B is 16.42 L
23. Which of the following is/are correct statement(s)?
 (A) If a solution is prepared by dissolving excess of AgNO_3 solution in KI solution, the charge likely to develop on colloidal particle is negative
 (B) As number of carbons increases in sodium alkyl sulphate surfactant in water, its kraft's temperature increases
 (C) The conductivity of a solution having surfactant molecules decreases sharply at the CMC
 (D) At high temperature N_2 is chemisorbed on the iron surface
24. Which of the following is/are capable of liberating SO_2 gas on reaction with hot and conc. H_2SO_4 ?
 (A) NaI
 (B) NaCl
 (C) Cu
 (D) S
25. A scarlet compound A is treated with conc. HNO_3 to give a chocolate brown precipitate B. The precipitate is filtered and the filtrate is neutralized with NaOH. Addition of KI to the resulting solution give a yellow precipitate C. The precipitate B on warming with conc. HNO_3 in the presence of $\text{Mn}(\text{NO}_3)_2$ produce a pink-coloured solution due to the formation of D. Correct statement(s) regarding A to D is/are
 (A) A is Fe_3O_4
 (B) B has a metal ion in +4 oxidation state
 (C) C is PbI_2
 (D) D is Mn_2O_7

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

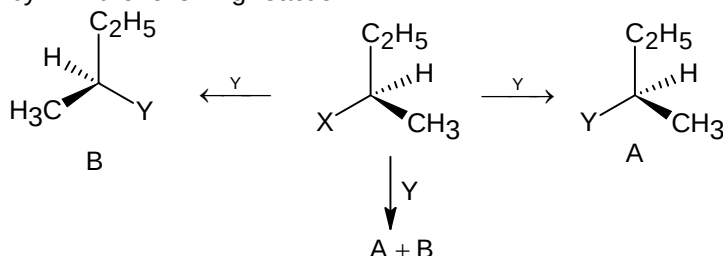
Paragraph for Question Nos. 26 and 27

In the permanganate titrations, KMnO_4 is used as an oxidizing agent in acidic medium. The medium is maintained acidic by the use of dilute H_2SO_4 . KMnO_4 acts as self indicator. Before the end point, the solution remains colourless but after the equivalence point only one extra drop of KMnO_4 solution imparts pink colour. These titrations are used for the estimation of ferrous salts, oxalic acid, oxalates, arsenites etc.

26. In acidic medium, As_2O_3 is converted to Arsenic acid by KMnO_4 . In this reaction, ratio of mols of KMnO_4 reacted to mols of Arsenic acid formed is
- (A) $\frac{4}{5}$
- (B) $\frac{4}{3}$
- (C) $\frac{2}{5}$
- (D) $\frac{4}{7}$
27. 200 ml of 0.01 M KMnO_4 oxidise 10 ml of H_2O_2 solution in acidic medium. Determine the maximum volume of O_2 which can be obtained from 20 ml of same H_2O_2 solution at NTP?
- (A) 28 ml
- (B) 56 ml
- (C) 112 ml
- (D) 224 ml

Paragraph for Question Nos. 28 and 29

There are three outcomes for a reaction at an asymmetric atom. Consider the replacement of a group X by Y in the following reaction



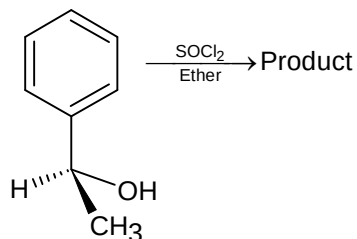
If (A) is the only compound obtained, the process is called retention of configuration.

If (B) is the only compound obtained, the process is called inversion of configuration.

If a 50 : 50 mixture of the above two is obtained then the process is called racemisation and the product is optically inactive, as one isomer will rotate light in the direction opposite to another.

In general, if during a reaction, no bond to the stereocentre is broken, the product will have the same general configuration of groups around the stereocentre as that of reactant. Such a reaction is said to proceed with retention of configuration.

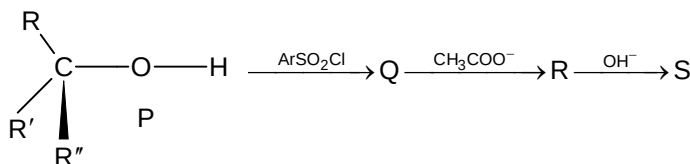
28.



Which is correct statement regarding the above reaction?

- (A) Kinetically, reaction is bimolecular
 (B) Stereo chemically, the configuration is retained
 (C) Stereo chemically Walden inversion takes place
 (D) In place of ether when pyridine used, we get a racemic mixture

29.

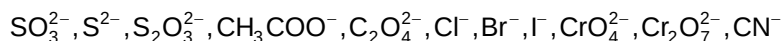


Which of the following statement is/are correct?

- (A) P and R must have the same configuration
 (B) S and R must have the same configuration
 (C) P and S must rotate plane polarized light to the same direction
 (D) Two among the 3 steps mentioned above involve inversion of configuration

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

 30. How many of the acid radicals on treatment with excess AgNO₃ gives white precipitate?

 31. How many of the following molecules involve d-orbital in π – bonding?

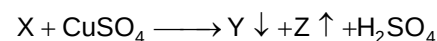
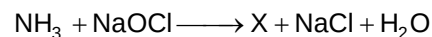

32. A current strength of 1 Amp is passed through 100 ml of 0.05 M KCl solution for 96.5 sec. After the electrolysis is complete, 20 ml of 0.1 M weak acid HA is added to the above solution. Determine pH of the final solution. [K_a of HA = 10^{-5} ; Assume no change in volume]

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

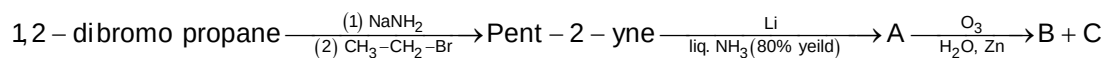
Question Stem



33. What is the weight of Y obtained; when 48 g 'X' is added to excess CuSO_4 . [Cu = 63.5; S = 32; O = 16; N = 14]
34. If the oxidation state of N in 'X' is a, then value of $\left| \frac{2a}{3} \right|$ is

Question Stem for Question Nos. 35 and 36

Question Stem



35. Weight of NaNH_2 required to convert one mole 1,2-dibromopropane is (Assume 100% conversion) x gm, then value of $\frac{x}{2}$ is [Na = 23, N = 14]
36. What would be weight of C (Assuming it contains less carbon than B)

Question Stem for Question Nos. 37 and 38

Question Stem

Metal A $\xrightarrow[4.25 \text{ eV}]{\text{Photons}}$ Maximum KE of ejected electron is T_A (in eV); and de-Broglie wave length = λ_A

Metal B $\xrightarrow[4.20 \text{ eV}]{\text{Photons}}$ Maximum KE of ejected electron is T_B (in eV); and de-Broglie wave length = λ_B

If $\lambda_B = 2\lambda_A$ and $T_B = T_A - 1.50 \text{ eV}$ then

37. Work function of A is _____
38. $T_B =$ _____

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. Let $I = \int_{\pi/6}^{\pi/3} \left(e^{2 \sin x} + e^{\frac{3 \operatorname{cosec} x}{2}} \right) \cot x \, dx$, then

(A) $2I = \int_{1/2}^{3/2} \left(\frac{e^{2x} + e^{\frac{3}{2x}}}{x} \right) dx$

(B) $4I = \int_{1/2}^{3/2} \left(\frac{e^{2x} + e^{\frac{3}{2x}}}{x} \right) dx$

(C) $I < \frac{2}{\sqrt{3}} (e^{\sqrt{3}} - e)$

(D) $I > \frac{2}{\sqrt{3}} (e^{\sqrt{3}} - e)$

40. If X and Y are two non-singular matrices such that $3X - 4YY^T = I$ and $Y^{-1} = 2X^T$ (where I represents identity matrix of order 3). If $\operatorname{Tr}(A)$ represents sum of diagonal elements of matrix A), then

(A) $|2Y^T - Y + 4Y^3 - 5YX| = -1$

(B) $\operatorname{Tr}((2X)^{-1} + I - 2XY + 4Y^3) = \frac{9}{2}$

(C) $|(2X)^{-1} + 4Y^3| = \frac{27}{8}$

(D) $\operatorname{Tr}(3X^3 - X^2) = 3$

41. In a knock out tournament, 16 equally skilled players $S_1, S_2, \dots, S_{16}$ are participating. In each round players are divided into pairs at random and winner from each pair moves in the next round, then choose the correct option(s)

(A) probability that S_2 reaches the semi-final is $\frac{1}{15}$

(B) probability that S_1 reaches the semi-final given that S_2 reaches the semi-final is $\frac{1}{5}$

(C) probability that S_1 wins the tournament given that S_2 reaches the semi-final is $\frac{1}{20}$

(D) probability that S_2 reaches the semi-final given that S_4 does not reaches the semi-final is $\frac{1}{3}$

42. If the equation $\sec\left(2\cos^{-1}\left(\tan\left(\cot^{-1}\left(\operatorname{cosec}\left(5\sec^{-1}x\right)\right)\right)\right)\right)=1$ has k number of solutions represented by x_i ; ($i = 1, 2, \dots, k$), then choose the correct option(s)
- (A) $k = 4, \sum_{i=1}^4 x_i = 2\pi$
- (B) x_i satisfies the equation $5x^4 - 20x^2 + 16 = 0$
- (C) x_i satisfies the equation $5x^4 - 20x^2 - 16 = 0$
- (D) $\sum_{i=1}^k \cos^{-1}\left(\frac{1}{x_i}\right) = 2\pi$
43. Let L_1 and L_2 be two lines passing through origin and making an angle $\frac{\pi}{3}$ with the line $L \equiv \frac{x-3}{2} = \frac{y-3}{\lambda} = \frac{z}{\lambda}$. If direction ratios of L_1 are proportional to $\frac{3}{2}, 3, -\frac{3}{2}$, then
- (A) equation of the plane containing L_1 and L_2 is $x - y - z = 0$
- (B) equation of plane containing line L and perpendicular to plane containing L_1 and L_2 is $y - z = 3$
- (C) the volume of tetrahedron having vertex $(1, 1, 1)$ and base formed by intersection of L_1, L_2 and L is $\frac{1}{2}$
- (D) the image of point $(1, 1, 1)$ about the plane containing L_1 and L_2 is $\left(\frac{5}{3}, \frac{1}{3}, -\frac{1}{3}\right)$
44. Let $f(n) = \sum_{r=1}^n \frac{4n+1}{r} C_{4r-1}$, then
- (A) $f(10) = (2^{40} - 1) \frac{2}{21}$
- (B) $f(20) = (2^{80} - 1) \frac{2}{21}$
- (C) $f(50) = (2^{200} - 1) \frac{2}{101}$
- (D) $f(n) = (2^{4n} - 1) \frac{2}{(n+1)}$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

Let $f(x)$ be a polynomial function satisfying $\int_0^x (f(t))^3 dt = \frac{32}{5x^2} \left(\int_0^x f(t) dt \right)^3 \quad \forall x \in \mathbb{R} - \{0\}$. Also $f(1) = 2$

45. If the line $y = m(x - 1)$; $m \neq 0$ touches the curve $y = f(x)$, then m is equal to
- (A) $\frac{27}{4}$
 (B) $\frac{27}{2}$
 (C) 27
 (D) $\frac{9}{2}$
46. If $g(x)$ be a function such that $\int_0^1 (f(x) - g(x))g(x)dx = \frac{1}{7}$, then the area bounded by the curve $y = g(x)$, $y = 2^{5-x}$ and x -axis is
- (A) 4
 (B) 8
 (C) $4 + \frac{8}{\ln 2}$
 (D) $8 + \frac{4}{\ln 2}$

Paragraph for Question Nos. 47 and 48

Consider two ellipse $S_1 : \frac{x^2}{a^2} + \frac{y^2}{4} = 1$; $S_2 : \frac{x^2}{15} + \frac{y^2}{b^2} = 1$; $4 < a^2 < b^2 < 15$.

Let a tangent is drawn to the curve S_1 at point P so as to intersect the curve S_2 at points Q and R . Also tangents to the curve S_2 at Q and R intersect at point $S(3, 4)$. Moreover the foot of perpendicular from one of the focus of S_1 on the tangent at P is $\left(1 + \frac{4}{\sqrt{5}}, 2 - \frac{2}{\sqrt{5}}\right)$

47. Point of contact P is
- (A) $\left(-\frac{9}{5}, \frac{17}{5}\right)$
 (B) $\left(\frac{9}{5}, \frac{8}{5}\right)$
 (C) $\left(-\frac{9}{5}, -\frac{8}{5}\right)$
 (D) $\left(\frac{9}{5}, -\frac{17}{5}\right)$
48. If normals to the curve S_2 are drawn at points Q and R so as to intersect at T , then coordinates of T are
- (A) $\left(\frac{4}{11}, \frac{13}{11}\right)$
 (B) $\left(\frac{3}{11}, \frac{4}{11}\right)$
 (C) $\left(-\frac{3}{11}, -\frac{4}{11}\right)$
 (D) $\left(\frac{15}{11}, \frac{10}{11}\right)$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. The number of solution(s) of the equation $2\sin\left(\frac{\pi}{4} + 25x\right) = \sqrt{1 + 8\sin 50x \cos^2 50x}$; $x \in \left(0, \frac{\pi}{25}\right)$ is/are
50. The equation $z^{100} + (15z - 1)^{100} = 0$ has 100 complex roots $z_i, \bar{z}_i$ ($i = 1, 2, 3, \dots, 50$), then the value of $\frac{1}{100} \left[\sum_{r=1}^{100} \frac{1}{|z_r|^2} \right]$ is
51. Let $I_n = \int_0^{\pi} \left(\frac{1 - \cos nx}{1 - \cos x} \right) dx$; $n \in \mathbb{I}$, then the value of $\left(\frac{I_{51} - I_{50} + I_{49}}{\pi} \right)$ is

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53**Question Stem**

Consider a linear pattern which consists of red and blue dots. Let x_n be the number of patterns of length 'n' that do not contain the sequence RR (R stands for red dot) and y_n be the number of patterns of length 'n' that do not contain the sequence BBB (B stands for blue dot)
(Example: BBRBRBBR is a pattern of length 8)

52. The value of y_7 is
53. The value of x_{10} is

Question Stem for Question Nos. 54 and 55**Question Stem**

Let $A = \begin{bmatrix} 3 & 0 & 0 \\ 6 & 3 & 0 \\ 9 & 6 & 3 \end{bmatrix}$ and $B = [b_{ij}]$ is a 3×3 matrix such that $B = \left(\frac{A}{3}\right)^{50} - I$ where I is identity matrix of order 3

54. The value of b_{31} is
55. The value of $\det(A \cdot \text{Adj } 2(B + I))$ is

Question Stem for Question Nos. 56 and 57

Question Stem

Consider a right angled triangle having integral sides a , b , and c ; ($a < b < c$). If sum of perpendicular sides differ the third side by 6, then

56. The maximum possible area of triangle is
57. If $|a - b|$ is minimum, then the length of angle bisector through $\angle C$